

Is the wall behaving as expected?

Grey-box monitoring of station 02 (2018–2026)

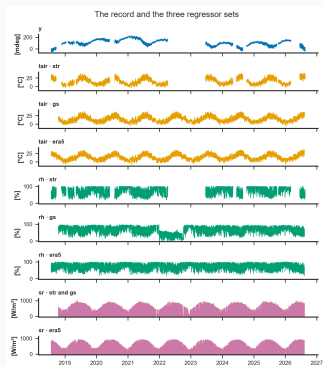
The Problem

Question: Given the current environment, is the wall leaning exactly as expected, or doing something unusual?

- Readings every 20 minutes from the medieval wall of Gubbio:
 - Inclination (leaning)
 - Air temperature
 - Humidity
 - Solar radiation
- **Goal:** Model normal behavior and monitor for unexpected movements over 8 years.

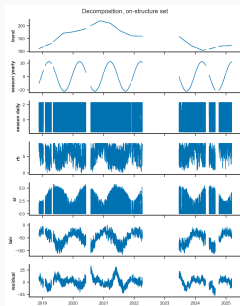
The Dataset

- We use three sets of environmental drivers to ensure reliability when the wall's own sensors (`str`) fail.
- External proxies: Local ground station (`gs`) and ERA5 reanalysis (`era5`).



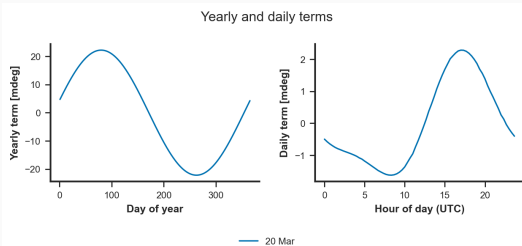
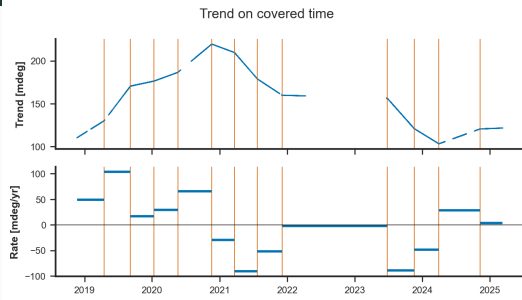
How We Model the Wall

- We use a “grey-box” model (*NeuralProphet*) separating inclination into:
 - **Trend:** Long-term wandering.
 - **Seasonality:** Annual and daily repeating cycles.
 - **Drivers:** Temperature, humidity, radiation.



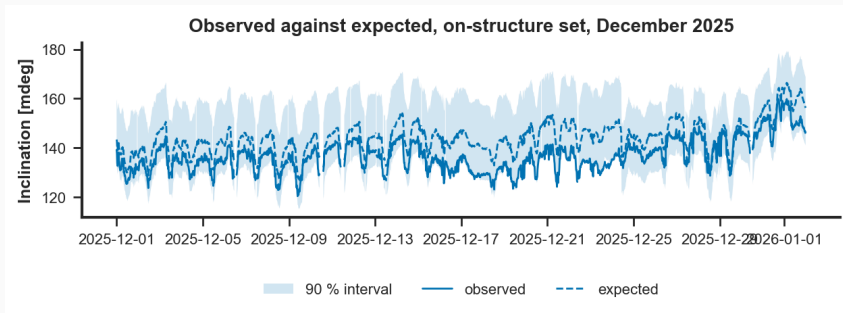
What Drives the Wall's Movement?

- **Slow Trend ($\approx 60\%$):** Major wandering over years. Not a straight drift; it reverses.
- **Air Temp (15–24%):** The wall and air heat together from solar radiation. The daily swing is captured instantaneously by air temperature.
- **Annual Cycle (12–17%):** Delayed seasonal response.



Monitoring for Anomalies

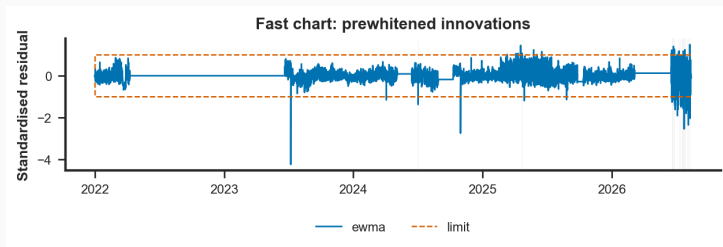
- **Rolling Expectation:** The model is re-trained every 30 days to establish a moving baseline.



The Control Charts

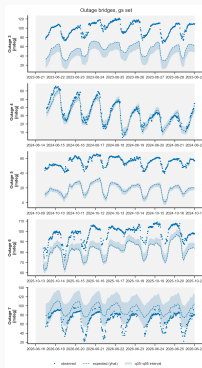
We use three control charts to catch different anomalies:

1. **Fast Chart:** Catches sudden steps or spikes (e.g., instrument faults).
2. **Daily Chart:** Catches changes in daily swing amplitude or phase.
3. **Slow Chart:** Catches slow continuous drift over weeks.



Behavior Across Outages

- What happens when sensors go offline? The monitor flags when the wall resumes at an unexpected level.



- The wall at station 02 acts largely as a thermometer, driven primarily by solar heating (captured via air temperature) and a slow, reversible structural drift.
- By separating this expected environmental response from raw data, our model and control charts provide a sensitive, automated way to monitor the true structural health.